

Automated Social Media Content Performance Analysis

Business Problem

A multi-platform brand posting regularly across 6 social media platforms had no consolidated view of what content actually drove performance. Post-level data covering 5,600 posts existed as a single flat, denormalised export with no analytical structure.

The business could not identify which content categories and post types drove engagement, whether posting time or platform choice meaningfully affected results, or where gaps in click tracking limited how confidently funnel decisions could be made.

Analytical Framework

A structured analytics solution was built using the SCAN Framework. The business problem was scoped through a full Python exploratory analysis, a metric hierarchy was defined with Average Engagement Rate as the North Star, and a Power BI report was built on a normalised star schema data model.

The report delivers five focused pages covering the North Star, the Volume, High Performer, and Click-Through Rate catalysts, and diagnostic breakdowns by platform, content category, post format, and region. Every visual traces back to the North Star.

Headline Results

Total Posts	Avg Engagement Rate	High Performer Rate	Top Engagement Rate
5,600	0.15	23.6%	0.20
<i>Across 6 platforms</i>	<i>Mean equals median</i>	<i>1,320 of 5,600 posts</i>	<i>Educational & Customer Story</i>

Key Results

Engagement Performance

Engagement Rate is the most stable metric in the dataset, with mean and median both at 0.15 and effectively zero skew. Platform choice barely moves the number, ranging only from 0.14 on LinkedIn to 0.16 on Instagram, confirming that what gets posted matters far more than where it gets posted.

Performance Tiers

Of all 5,600 posts, 23.6% (1,320) qualify as High engagement performers, 55.5% sit in the Medium tier, and 20.9% fall into Low. PDF posts reach the High tier most often at 44%, followed by Image posts at 36%, both well above the portfolio average.

Content and Format

Content category shows the widest performance spread of any dimension. Educational and Customer Story content both post a 0.20 median engagement rate, well above Entertainment at 0.08. Educational is the most produced category at 36.2% of posts, while Video is the most produced format at 52.6% despite a lower 0.14 median engagement rate than Image or PDF at 0.18.

Click Tracking

Click data is only available for 3 of 6 platforms. Facebook, LinkedIn, and TikTok report a consistent 2% click-through rate, while Instagram, X.com, and YouTube return none at all, leaving 3,740 of 5,600 posts (66.8%) without a CTR figure.

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Recommendations and Business Impact

The following recommendations are drawn directly from the dataset. Each is paired with the specific business impact it addresses, quantified where the data supports it.

#	Recommendation	Business Impact
1	Reallocate content production toward Educational and Customer Story formats, and reduce investment in Entertainment content.	✓ Educational and Customer Story content post a 0.20 median engagement rate versus 0.08 for Entertainment, a gap of 150%. ✓ Lifting underperforming category output toward the 0.20 benchmark raises overall engagement without increasing total posting volume.
2	Rebalance the content mix toward Image and PDF formats where current strategy over-indexes on Video.	✓ Image and PDF posts achieve a 0.18 median engagement rate versus 0.14 for Video, a 29% improvement per post shifted. ✓ Shifting 10% of the 2,946 Video posts to Image or PDF format would convert approximately 295 posts to a higher-performing format without increasing total posting volume.
3	Deprioritise best-time-to-post optimisation and redirect that planning effort into content category and format strategy.	✓ Platform choice moves engagement rate by just 0.02 across all 6 platforms, while content category spans 0.12 (0.08 to 0.20), making category 6x the lever that platform is. ✓ Redirecting time spent on posting schedule optimisation toward content category strategy targets the variable with the largest measurable impact.
4	Audit the Medium-tier majority for the content and format attributes shared with the High-tier 23.6%, and apply them systematically.	✓ PDF posts already achieve a 44% High engagement rate and Image posts 36%, both well above the 23.6% portfolio average. ✓ Moving 10% of the 3,108 Medium-tier posts into the High tier would add approximately 311 High-performer posts, lifting the High Performer Rate from 23.6% to 29.2%.
5	Prioritise closing the click-tracking gap on Instagram, X.com, and YouTube before using CTR in funnel decisions.	✓ 3,740 of 5,600 posts (66.8%) have no click data at all, making CTR an unreliable cross-platform metric in its current state. ✓ Closing this gap would unlock a genuinely comparable six-platform CTR metric for future funnel reporting.